

# Garbage in, garbage out: stringent quality control of behavioral data boosts signal in brain–behavior associations

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Low reproducibility in brain–behavior association studies has prompted calls for larger samples, but rigorous quality control (QC) of behavioral measures may also strengthen the robustness of observed associations. This study examined how varying levels of behavioral QC affect brain–behavior associations using delay discounting, a widely studied measure of immediate reward orientation. Data were drawn from the Adolescent Brain and Cognitive Development Study ( $n = 10,936$ ), and delay discounting data were subjected to eight increasingly stringent QC levels. Associations between delay discounting and brain structural morphometry were evaluated in 15 bilateral a priori regions of interest (ROIs), and changes in effect size, significant associations, and multivariate behavioral prediction were quantified across QC levels. Greater QC stringency substantially amplified association strength, with effect sizes increasing by an average of 188.52% across ROIs, the number of statistically significant associations more than doubling, and multivariate prediction improving by 267%. Although higher QC levels reduced sample size by up to 65%, strengthening of associations plateaued at intermediate levels, suggesting asymptotic improvements in resolution. Overall, stringent behavioral QC substantially increased the magnitude of brain–behavior associations, suggesting this is a critical methodological consideration for improving reproducibility in cognitive neuroscience.

**Keywords** behavioral neural correlates, brain–behavior associations, delayed reward discounting, impulsivity, quality control

## Introduction

For more than a decade, cognitive neuroscience has been grappling with low reproducibility of research findings, the so-called replication crisis (Barch and Yarkoni 2013; Poldrack et al. 2017; Kelly and Hoptman 2022). This is particularly the case for brain–behavior association studies, which examine features of the brain, either structural or functional, in relation to independent psychological or psychiatric phenomena (Masouleh et al. 2019; Marek et al. 2022). One initial explanation postulated for the lack of replication is that true effect sizes are in fact smaller than initially expected and early studies were thus underpowered to detect relations reliably (Button et al. 2013; Marek et al. 2022; Liu et al. 2023). These observations have led to the collection of large-scale multisite “open science” datasets (Szucs and Ioannidis 2020). Notably, however, “big data” are also accompanied by additional challenges (Smith and Nichols 2018). If, for example, error variance “noise” outpaces true effect “signal” due to site implementation differences, larger samples alone will not produce more robust findings.

Strategies for addressing reproducibility typically prioritize improving neuroimaging methods (Barch and Yarkoni 2013; Dosenbach et al. 2017; Fair et al. 2020; Kelly and Hoptman 2022; Kang et al. 2024), but a

complementary underdiscussed strategy is to maximize the resolution of the psychological phenomena. Many of the processes investigated in brain–behavior association studies are performance-based indicators, such as behavioral tasks or neuropsychological assessments, with highly variable data quality in terms of individual participants' comprehension, attention, and effort. In other words, some participants may not understand a task adequately, pay full attention while completing the task, or give their optimal performance on the task, diluting the characterization of the phenomenon of interest. The challenge posed by poor-quality data are further amplified in large multisite studies where there may be more variability in how the tasks are administered across sites and where researchers outside project team using the data may not fully understand how the task is administered or be aware of the necessary nuances in quality control (QC). Furthermore, the potential for low-quality data further compounds in studies using behavioral tasks in children or adolescents, who are more susceptible to issues of comprehension, attention, effort, and may be more likely to fundamentally not understand the task. From this perspective, careful systematic QC of psychological and neurocognitive processes is essential for robust detection of associated brain features.

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The current study uses delay discounting (DD) as a prototypic example to illustrate the critical role of QC in brain–behavior relations. DD is a behavioral economic measure of intertemporal decision-making that putatively reflects a form of impulsivity (Stahl et al. 2013; Bickel and Marsch 2001). Operationally, DD characterizes how much an individual devalues a reward (typically financial) based on its delay in time, thus reflecting a person's orientation toward immediate gratification. These now-vs.-later preferences represent a fundamental form of human decision-making orientation, varying as a normative trait within the population (Odum 2011), but with high valuation of immediate rewards being trans-diagnostically associated with a wide variety of health outcomes associated with deficits in self-regulation and overconsumption, such as addiction, obesity, and ADHD (MacKillop et al. 2011; Jackson and Mackillop 2016; Amlung et al. 2017, 2019). In turn, understanding the neural underpinnings of DD is essential for elucidating the brain mechanisms that link DD to adverse health outcomes.

Previous studies on the neural correlates of DD have implicated the lateral prefrontal cortex (PFC), which is well-established for its role in executive function and self-control, and the superior frontal gyrus, which is also implicated in cognitive control processes (Bjork et al. 2009; Schwartz et al. 2010; Cho et al. 2013). Also implicated, the medial PFC and anterior cingulate cortex are central to conflict monitoring and decision-making, while the putamen plays a role in reward processing and impulsivity (Dombrowski et al. 2012; Bernhardt et al. 2014). The involvement of the middle temporal gyrus and entorhinal cortex underscores the role of memory and contextual information in decision-making (Owens et al. 2017). While most of these findings are based on studies on adults, evidence suggests substantial overlap in the neural substrates of DD between adults and adolescents (Wang et al. 2017; Bounoua et al. 2023).

As with many brain–behavior associations, however, DD findings have also been inconsistent across studies (Pashler and Wagenmakers 2012; Baker and Penny 2016; Poldrack et al. 2017; Masouleh et al. 2019; Garzón et al. 2022; Elsayed et al. 2024), which may stem from variability in QC the DD decision making phenotype. Procedurally, DD tasks typically present participants with a randomized array of dichotomous choices between smaller immediate rewards and larger delayed rewards with highly variable amounts and durations. Although ostensibly simple, participants may nonetheless not understand the task, stop paying attention over the choice trials, or simply not give good effort overall, akin to other decision-making tasks or executive function tasks. Furthermore, DD decision-making exhibits substantial developmental changes during adolescence (Chen et al. 2021; Klein et al. 2022), based on both dramatic changes in brain development and cognitive understanding of the task contingencies. Collectively, robust QC of DD is essential for a high-resolution psychological phenotype and, in turn, to derive meaningful and reproducible associations with brain morphometry, and even greater for adolescents (Anokhin et al. 2014).

Thus, using DD as an exemplar cognitive phenotype, the current study aims to: (i) demonstrate the importance and impact of QC on associations with brain morphometry, testing the hypothesis that higher-quality DD data will strengthen these associations; (ii) provide collateral validation of the QC approach by testing whether it similarly amplifies a well-established behavioral correlate of financial DD, namely income (MacKillop and Kahler 2009; MacKillop et al. 2010; Assari et al. 2018; Peviani et al. 2024); and (iii) replicate and confirm previously reported associations between DD and brain morphometry, including their directionality. These aims are undertaken in the

large cohort of adolescents from the Adolescent Brain Cognitive Development (ABCD) study (Casey et al. 2018), a large-scale multisite study on adolescents that has both a large sample size for high statistical power but a design that is also subject to increased threats to data quality (ie multiple sites, a complex behavioral paradigm). Collectively, these objectives sought to demonstrate the signal boost yielded by rigorous QC of behavioral measures in brain–behavior association studies toward improving robustness in cognitive neuroscience.

## Materials and methods

### Participants

The ABCD Study is an ongoing longitudinal research project designed to collect structural and functional brain scans, along with behavioral, neuropsychological, and psychosocial data, from nearly 12,000 youths across a 10-year period spanning adolescence. In the present study, we focused exclusively on the first wave of brain morphometry data (2016–2018; ages 9 to 10) and DD data from the first follow-up visit (2017–2019; ages 10 to 11). Participants with both baseline brain morphometry and DD data were included ( $n=10,936$ ). Additional demographic details and information on the sample included are provided in Table S1.

### Delay reward discounting task

In the ABCD study, a 42-item adjusting-amount DD task is used (7 choices for 6 times) was employed (Delay Discounting Task—Millisecond: [https://www.millisecond.com/download/library/v6/delay\\_discountingtask](https://www.millisecond.com/download/library/v6/delay_discountingtask)). This task required participants to choose between an immediate smaller reward and a delayed larger reward of \$100, available at varying intervals (6 hours, 1 day, 1 week, 1 month, 3 months, 1 year, and 5 years). Participants' choices were used to determine indifference points, which represent the subjective value at which participants found the immediate and delayed rewards equally preferable. These indifference points were then used to calculate the *log* transform (of base 10) of the discounting rate ( $k$ ) using Mazur's equation (Mazur 1987), as well as the area under the curve (*AUC*). However, since the traditional *AUC* measure tends to overemphasize indifference points at longer delays while underrepresenting those at shorter delays, we included a modified *AUC* measure, *AUClogD*, following the approach suggested by Borges et al. (2016). This paper primarily focuses on results based on *log(k)* to maintain consistency with the hyperbolic regression model used for QC. However, to facilitate comparisons with other studies and ensure comprehensiveness, all *AUClogD* results are provided in the supplementary materials. Another reason for including these results is that *AUC*, as an atheoretical measure, has been suggested to be preferable for assessing impulsivity in children and adolescents (Burns et al. 2021; Klein et al. 2022).

We implemented two types of QC pipelines for DD. The first type focused on the goodness of fit of the indifference points to a nonlinear hyperbolic regression model, a widely used approach in modeling DD behavior (Cheng and González-Vallejo 2014). In this context, hyperbolic regression models the relationship between the value of a reward and the delay in receiving it, as described in Equation 1 (Mazur 1987):

$$V = \frac{A}{1 + kD} \quad (1)$$

where  $V$  is the present value of the reward,  $A$  is the reward amount,  $D$  is the delay, and  $k$  is the discounting parameter. A steeper discounting

curve, indicated by a higher  $k$  value, reflects a stronger preference for immediate rewards. To ensure data quality and assess the goodness of fit for the hyperbolic model to participants' discounting behavior, we applied 7 hyperbolic regression fitness measure  $R^2$  cut-off scores (i.e.,  $R^2 \geq 0.1, 0.2, 0.3, 0.4, 0.5, 0.6$ , and  $0.7$ ), with higher cut-off scores corresponding to a more stringent QC and more data points removed. Figure 1 shows a visualization of the investigated 7  $R^2$  cut-off scores of hyperbolic regression. Of note, several other equations have been used to model DD data (Gilroy et al. 2017), but the hyperbolic model is the most commonly used (MacKillop et al. 2011).

The second type was based on the criteria proposed by Johnson and Bickel (2008; J&B), which provide a model-independent heuristic-based method for identifying nonsystematic DD data. According to their algorithm, an individual's DD data is considered invalid if it meets either of the two rules: (i) if a later indifference point exceeds an earlier one by more than 20% of the larger reward, or (ii) if the difference between the first and last indifference points is less than 10% of the larger reward (\$100). These heuristic criteria were designed to filter out data that do not conform to basic expectations of DD decision making.

By applying both types of QC pipelines, 8 cleaned datasets were generated: one using the J&B rules and 7 others based on the hyperbolic regression model cut-off scores (H 0.1, H 0.2, H 0.3, H 0.4, H 0.5, H 0.6, and H 0.7) for the hyperbolic fit. These filtered datasets were then compared to determine which provided stronger associations with the brain morphometry in terms of standardized effect sizes and the adjusted  $R^2$  from the linear regression model.

## Imaging acquisition and preprocessing

Structural imaging data were acquired using T1-weighted (T1w) scans across 21 data collection sites, utilizing three different 3T scanners. The T1w acquisition, with an isotropic 1 mm resolution, employed a 3D T1w inversion-prepared RF-spoiled gradient echo sequence (Casey et al. 2018). Prospective motion correction was applied where available, specifically on Siemens and GE scanners (White et al. 2010; Tisdall et al. 2012). For details on the scanners' parameters, refer to Table S2. The T1w images underwent a series of preprocessing steps to ensure high data quality. Initially, gradient nonlinearity distortions were corrected using scanner-specific transformations provided by MRI manufacturers (Jovicich et al. 2006). Following this, intensity inhomogeneity was addressed by applying B1-bias field corrections, which normalized intensity variations across the images. The images were then rigidly registered and resampled into alignment with an average reference brain in standard space, facilitating standardized analysis (Friston et al. 1995). Cortical surface reconstruction and subcortical segmentation were conducted using FreeSurfer v5.3, a tool validated for use in pediatric populations and widely utilized in large-scale studies (Fischl et al. 2002; Fischl et al. 2004). For cortical parcellation, the Desikan atlas was employed, providing anatomical labelling of cortical regions (Desikan et al. 2006), while the subcortical segmentation was done using the ASEG atlas (Sederevičius et al. 2021). Both quality assurance and processing of the imaging data were done by the ABCD study team (Hagler et al. 2019).

The processed T1w images had undergone thorough MRI scan QC procedures implemented by the ABCD study team. Trained technicians visually inspected the cortical surface reconstructions, assessing for common imaging artifacts such as motion, intensity inhomogeneity, and inaccuracies in segmentation. Each image was assigned a QC

score, with severe artifacts leading to exclusion from further analysis. This MRI QC process ensured that only high-quality images were used in subsequent analyses, thereby enhancing the validity of the study's findings (Hagler et al. 2019).

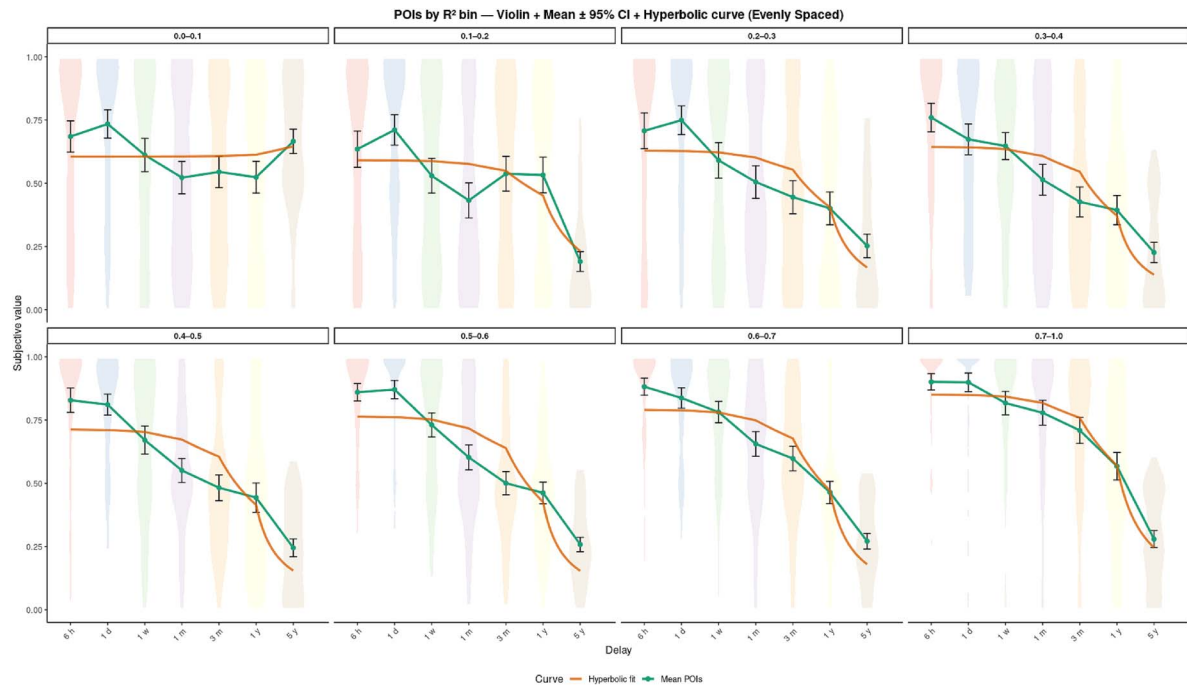
## Region of interest selection

The a priori regions of interest (ROIs) were selected based findings from large-sample quality-controlled morphometry brain-behavior studies of DD (Owens et al. 2019; Owens et al. 2020; Owens et al. 2021) in the Desikan atlas (Desikan et al. 2006). Key areas include the lateral PFC and superior frontal gyrus, identified for their involvement in evaluating future rewards (Bjork et al. 2009; Schwartz et al. 2010; Cho et al. 2013). The medial PFC, anterior cingulate cortex, and putamen were also included, as they have been implicated in decision-making and impulsivity (Dombrovski et al. 2012; Bernhardt et al. 2014). Additionally, the posterior cingulate cortex, frontal pole, middle frontal gyrus, caudate, middle temporal gyrus, and entorhinal cortex were selected based on their relevance to goal-directed behavior and memory (Monterosso et al. 2007; Tschernegg et al. 2015; Wang et al. 2016; Owens et al. 2017). All selected ROIs were aligned with the corresponding regions in the Desikan atlas, resulting in 30 ROIs but because of very high hemispheric correlation (mean  $r = 0.6, P < 0.0001$ ; see Fig. S1; Hageraves et al. 2025), all regions were then averaged bilaterally, for a total of 15 ROIs.

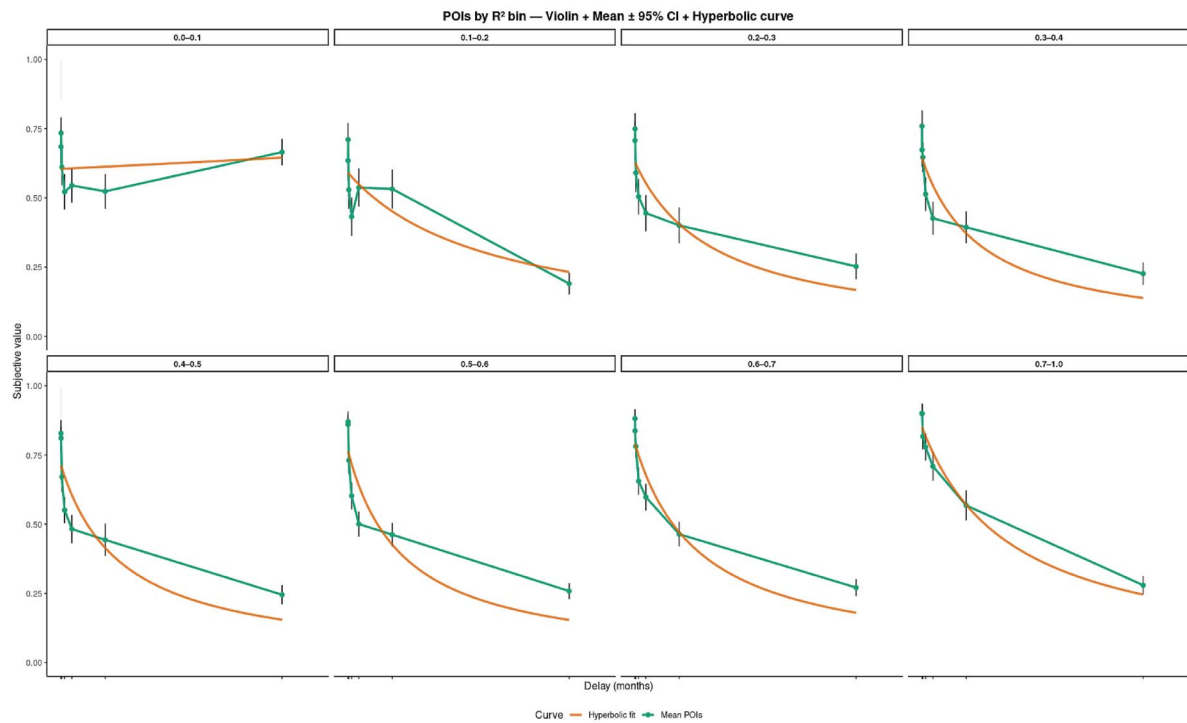
## Statistical analysis

The principal analyses assessed the strength of association between each ROI and the DD across the 9 datasets generated by the different QC pipelines (ie raw unfiltered, filtered using the J&B rules, and filtered with seven different  $R^2$  cut-off values in the hyperbolic model: H 0.1, H 0.2, H 0.3, H 0.4, H 0.5, H 0.6, and H 0.7). This analysis utilized brain morphometry data (independent variable) in relation to the base-10 logarithm of the discounting rate  $\log(k)$  (dependent variable). To examine the relationship between each ROI volume and DD, a linear regression model estimated using ordinary least squares was employed, adjusting for parental income, child's sex assigned at birth, and intracranial volume. Multiple comparisons were accounted for by controlling the false discovery rate (FDR) (Benjamini and Hochberg 1995). An ROI was deemed significantly associated with DD if the FDR-corrected  $P$ -value was  $< 0.05$ . To assess the influence of QC pipelines on ROI-DD associations, we compared standardized regression coefficients (ie beta weights) from linear models using the different generated QC datasets. To quantify the impact of QC pipelines, the percentage improvement in standardized regression coefficients for each region in each QC pipeline was calculated relative to the unfiltered data. For all investigated ROIs, beta weights obtained from the filtered-data regression models were compared to those from the unfiltered-data regression model using two-tailed paired  $t$ -tests. Subsequently, all individual ROIs were subsequently entered into a multiple regression model to further quantify the relationship between overall brain morphometry and DD. Difference in the adjusted  $R^2$  values was compared across the 8 different QC filtered datasets as well as the unfiltered dataset to identify the pipeline that yielded the highest variance explained while accounting for statistical significance (a function of sample size).

Finally, a zero-order correlation analysis was conducted to examine the relationship between annual parental income and DD behavior in



(a)



(b)

**Figure 1** Mean subjective value (points of indifference) across delay periods by  $R^2$  bin and their hyperbolic fit. Note. Panel (a) shows points of indifference (POIs) with evenly spaced delays for visual clarity, including violin plots displaying the distribution of individual responses. The shape of the hyperbolic function appears distorted because all delay points were forced to be equally spaced. Panel (b) presents the same data on a linear months scale to reveal the true hyperbolic discounting relationship, with error bars representing 95% confidence intervals around mean POIs. Both panels show mean POIs (green lines) and hyperbolic curve fits (orange lines) for each  $R^2$  bin (0.0–0.1, 0.1–0.2, 0.2–0.3, 0.3–0.4, 0.4–0.5, 0.5–0.6, 0.6–0.7, 0.7–1.0), where  $R^2$  indicates the goodness of fit of the hyperbolic discounting model. Delays range from 6 hours to 5 years.

both the filtered datasets produced by the different QC pipelines and the raw, unfiltered dataset. This analysis served as an independent, collateral validation of the QC approach. Parental income is a well-established and logical correlate of DD, with lower socioeconomic status associated with higher impulsivity (ie greater DD) (MacKillop and Kahler 2009; MacKillop et al. 2010; Assari et al. 2018; Peviani et al. 2024). It is logical insofar as individuals with fewer resources in the present would be expected to both be more sensitive to the value of an immediate reward and less comfortable deferring a financial outcome into the future. Therefore, improvements in the strength and consistency of this association following QC would provide face-valid evidence that the applied QC procedures enhance the signal quality of the behavioral data, beyond brain–behavior associations alone.

Although  $\log(k)$  was the primary DD index, sensitivity analyses were also performed using *AUClogD* (Borges et al. 2016) and the lateralized regions by hemisphere to assess whether any substantive differences emerged. All statistical analyses were conducted in R, Version 4.3.1.

## Results

### Univariate models

Using raw data, significant (FDR-corrected) inverse associations were present for 33% of ROIs in relation to DD, reflecting greater gray matter volume being associated with lower devaluation of immediate rewards. By the optimal QC stringency level ( $R^2 > 0.7$ ), 93% of the ROIs exhibited statistically significant associations, see Fig. 2. Magnitude of the observed standardized beta weights for each bilateral ROI in relation to DD  $\log(k)$  grew substantially across the filtered datasets (Fig. 2 and Fig. 3). Contrasts between the beta weights revealed that, compared to unfiltered data, the associations were significantly higher for all QC levels, and with a number of significant differences between the different QC levels of stringency, see Table S12. The univariate linear regression of the unfiltered data and the results for the other 8 different QC pipeline are shown Tables S3 to S11.

Compared to the unfiltered data, the mean percentage improvement in effect sizes across brain regions was 139.27% for J&B, with increasing improvements for the hyperbolic models from 155.17% for H-0.1 to 220.03% for H-0.7. Necessarily, the sample size decreased for all QC levels. Specifically, it was reduced by 68.03% for J&B—the most stringent in terms of number of samples removed, while the hyperbolic models reduced by 38.29% for H 0.1 to 61.77% for H 0.7. Figures 2 and S2 illustrate the percentage improvements in  $\beta$  weights for each brain region and percentage reduction of the sample size under different filtering methods, relative to the raw data. Of note, Table S14 shows the proportion of dataset filtered using the model-independent pipeline (J&B) that falls into each of the hyperbolic  $R^2$  cut-off ranges, suggesting over 72.25% of the data points retained using J&B fall into the highest quality threshold according to the most stringent hyperbolic model (H-0.7).

Sensitivity analyses for *AUClogD* and unilateral ROIs revealed the same patterns of results. See Tables S16 to S24 for the unilateral ROIs and Figs S3 to S5 and Tables S25 to S34 for *AUClogD* results.

### Multivariate models

In multivariate models, the unfiltered ROIs collectively explained less than 1% of the DD  $\log(k)$  variance. For datasets filtered using the hyperbolic regression model fit, the variance explained increased to up to ~3% at the highest level of stringency (H 0.7), see Fig. 3, an

approximately 2-fold increase. Filtering the dataset using the J&B pipeline resulted in intermediate improved variance explained of 1.85%.

In sensitivity analyses (Fig. S6 and Table S35), the multiple regression model was likewise able to explain substantially higher variance using the DD *AUClogD* as an outcome (raw = 1.65%, maximum QC-corrected = 4.09%).

### Relations with income

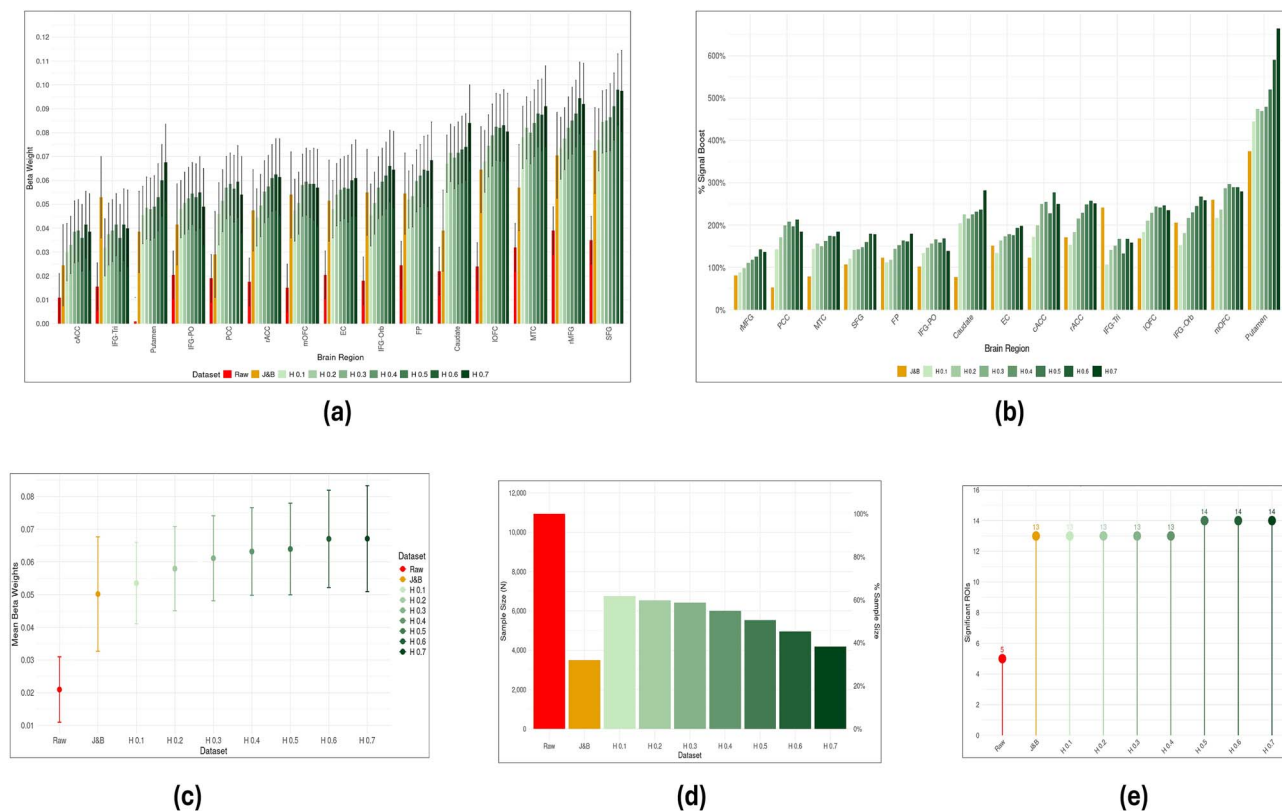
The same pattern of substantially higher  $\beta$  coefficients was present associations between child DD and parental income, with the correlation coefficient lowest at the unfiltered data ( $r = -0.024$ ) and highest at maximum QC  $R^2 = 0.7$ ,  $r = -0.123$ , shown in Fig. 3 and Table S13.

## Discussion

This study demonstrates the substantial impact of QC on the relationship between a psychological process, preference for smaller immediate rewards versus larger delayed rewards, and brain morphometry using a large open-science neuroimaging dataset. The analysis presented provides a systematic empirical demonstration that data quality is of paramount importance and should be prioritized alongside sample size when investigating the neural substrates of behavioral measures. Of note, the percentage improvement in standardized effect sizes varied considerably across brain regions, with notably greater increases observed in subcortical areas such as the putamen. This suggests that associations involving these subcortical may be particularly subtle and require higher data quality to be detected reliably. Additionally, it was notable that even directionality in the raw data was inconsistent in some regions, a pattern that was not present after filtering, indicating that such inconsistencies in the literature may stem from inadequate QC.

While this study offers a robust demonstration of how careful QC brings brain–behavior associations into sharper relief and concurrent insights into the neural correlates of DD in adolescents, a number of considerations should be noted. First, our findings further validate the notion that brain–behavior associations, particularly those linking brain morphometry to complex cognitive constructs such as DD, tend to be small in magnitude (Gratton et al. 2022; Gilman et al. 2024). Even with robust modeling approaches, these findings provide further evidence that only a modest proportion of variance in psychological processes can be explained by brain morphometry. Functional measures of brain activity, such as task-elicited neural activity during DD fMRI paradigms or resting-state connectivity would be expected to be more robust correlates of the behavior (Owens et al. 2018), but direct comparisons have not been made and this is necessarily conjecture. Integrating these functional metrics alongside structural features may offer a comprehensive understanding of the neurobiological underpinnings of DD and any neurocognitive process. Second, the data collection involved assessment asynchrony, with the structural MRI scan conducted prior to the DD task, almost certainly attenuating associations. Finally, this study used one illustrative paradigm, DD, to empirically evaluate the impact of QC and although the principle of the importance of carefully evaluating data quality is broadly applicable, the specific pipelines and cut-offs should not be assumed to apply to other cognitive paradigms.

An important consideration of the present approach is the trade-off between behavioral data validity and sample representativeness. By

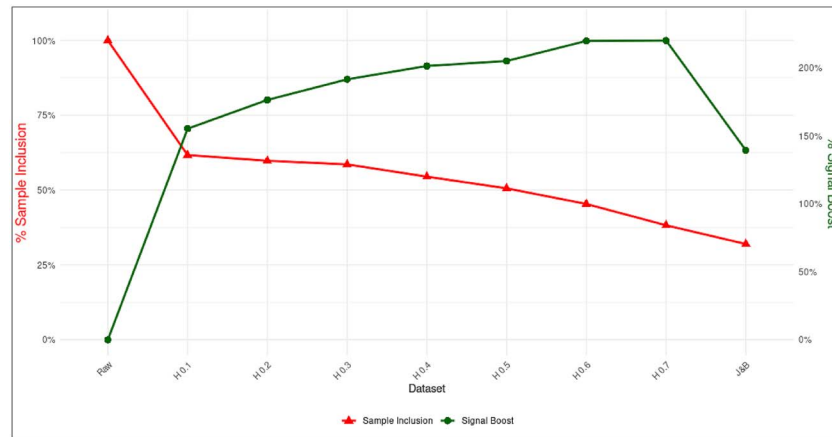


**Figure 2** A comparison of the associations for 15 brain gray matter volume ROIs in relation to DD across eight levels of QC stringency in univariate analyses. Panel (a) displays the magnitude standardized beta weights for the brain regions analyzed across datasets and their standard errors; panel (b) percentage improvements in the beta weights of brain regions across different filtering methods compared to raw, unfiltered data; panel (c) shows the mean standardized beta weights and SEMs across all bilateral brain regions; panel (d) sample size across datasets; panel (e) the number of significant regions out of the 15 bilateral brain regions across the datasets. Note. The dataset labeled H refers to data filtered using the hyperbolic regression model, where the number following H indicates the  $R^2$  cut-off value. J&B refers to the dataset filtered according to the Johnson and Bickel rules. Abbreviations for brain regions are as follows: cACC for caudal anterior cingulate, EC for entorhinal cortex, IOFC for lateral orbitofrontal cortex, mOFC for medial orbitofrontal cortex, MTC for medial temporal cortex, IFG-PO for inferior frontal gyrus pars opercularis, IFG-orb for inferior frontal gyrus pars orbitalis, IFG-tri for inferior frontal gyrus pars triangularis, PCC for posterior cingulate cortex, rACC for rostral anterior cingulate cortex, rMFG for rostral middle frontal gyrus, SFG for superior frontal gyrus, and FP for frontal pole.

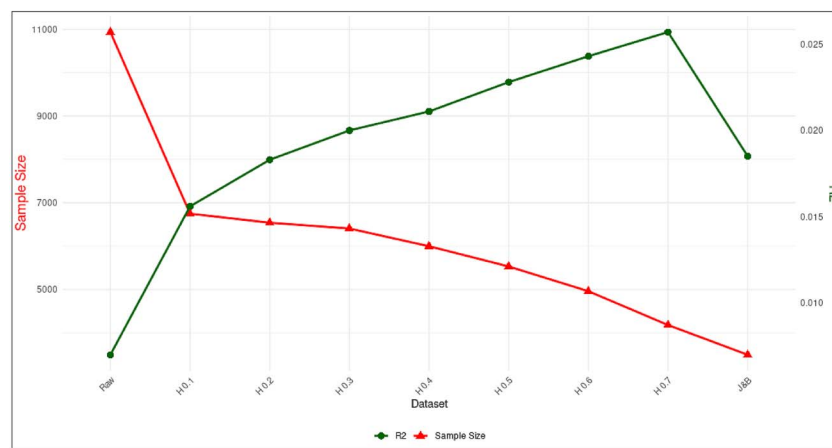
restricting analyses to participants who demonstrated adequate task comprehension and internally consistent responding, we improved the interpretability and reliability of brain–behavior associations; however, this comes at the potential cost of reduced generalizability (Gelino et al. 2025; Gelino et al. 2026). The ABCD study was designed to capture broad demographic, cognitive, and socio-environmental diversity, and exclusion based on behavioral validity may disproportionately affect certain subgroups, for example individuals with lower cognitive performance, reduced task engagement. Such exclusions may therefore introduce systematic bias and should not be interpreted as identifying “poor-quality participants”, but rather as reflecting limitations of task comprehension, measurement sensitivity, and developmental variability. We emphasize that stringent behavioral QC is intended to prevent biased or uninterpretable inferences, not to optimize effect sizes, and should be accompanied by transparent reporting of exclusion criteria and their impact on sample composition. Future work should explicitly examine how task design can be improved to preserve both measurement fidelity and inclusiveness.

In terms of DD-specific findings, the hyperbolic QC pipelines achieved larger standardized effect sizes than both the J&B and the raw data without any QC, with minimal sample removal. As a result, we propose a tiered framework for model fit thresholds based

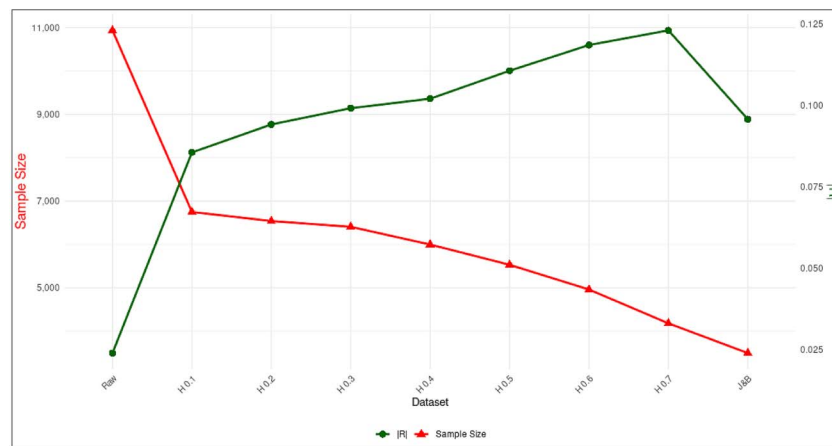
on the balance between effect size and data retention. Specifically, cut-offs in the range of 0.1–0.2 can be considered “liberal”, as they retain most of the sample while still yielding improved results compared to unfiltered data. Cut-offs between 0.3 and 0.5 represent a “moderate” level of strictness, showing stronger associations without substantial data loss. Finally, thresholds of  $\geq 0.6$  represent a “conservative” approach, aligning with stringent practices in behavioral economics demand curve research, though at the cost of greater data exclusion and thus lower statistical confidence. However, based on the multivariate results, an  $R^2$  of 0.3 appeared to achieve an optimal balance between sample inclusion and gain in explanatory power for both  $AUC_{logD}$  and  $\log(k)$ , a threshold that has also been applied in clinical samples (Reynolds and Schiffbauer 2004; Reynolds 2006). The challenges of the model-free J&B QC pipeline was evident in several ways. It excluded the largest proportion of participants but did not detect appreciably larger effect sizes. Furthermore, it included 22% of the sample who achieved a hyperbolic  $R^2$  value below 0.1, exhibiting very low model fit, while excluding 27.66% of the sample who achieved a hyperbolic  $R^2$  value of  $\geq 0.7$ , exhibiting very high model fit. This suggests the pipeline did not effectively filter out invalid responses. One possible explanation is that certain behavioral patterns may be subtle enough to violate the J&B heuristics while



(a)



(b)



(c)

**Figure 3** Coefficients of determination ( $r^2$ ) in multivariate models and associations with income in relation to sample sizes produced by the different QC pipelines. Panel (a) depicts mean percentage improvements in the standardized beta weights (signal boost) of all investigated brain regions and the percentage inclusion of the sample size across different filtering methods compared to raw, unfiltered data; panel (b) depicts total model variance accounted for in behavioral DD in the multivariate regression; panel (c) depicts  $|r|$  of the zero-order correlation between  $\log(k)$  and parental income. Note. H refers to the dataset filtered using the hyperbolic regression model, with the cut-off value for  $R^2$  specified by the number that follows H and J&B: Johnson and Bickel rules. See [Table S21](#) for the tabular presentation of the results.

still reflecting valid responses, or vice versa. These findings raise important concerns regarding the use of this pipeline, particularly in adolescent populations.

More broadly, although recent work has cautioned against relying solely on model-based  $R^2$  cut-offs for QC in behavioral economic tasks, (Gelino et al. 2024 Sep 1), due to potential bias and theoretical misalignment, our approach addresses these concerns in several ways. Specifically, we incorporated both a model-free pipeline (J&B) and a deliberately liberal  $R^2$  threshold ( $R^2 > 0.1$ ), supported by empirical findings demonstrating the appropriateness of this approach. Together, these steps reinforce the central thesis of our study: that rigorous yet balanced QC is essential to maximizing the detection of meaningful brain–behavior associations. Furthermore, the sensitivity analyses on *AUClogD*, a somewhat less widely used DD metric than *log(k)*, converged with the primary analyses and in fact had slightly higher magnitude of associations. This higher magnitude may be attributed to the fact that AUC, as an atheoretical measure, has been suggested to be more suitable for assessing impulsivity in children and adolescents (Burns et al. 2021; Klein et al. 2022).

This work suggests that inconsistent associations in brain–behavior associations for DD, and potentially numerous psychological processes, may at least partially be a consequence of inadequate or uncritical QC practices. In turn, more rigorous QC would be expected to capture more true brain–behavior associations and substantially improve reproducibility. In essence, our findings advocate for the adoption of standardized QC protocols as a crucial step toward more robust and reliable brain–behavior association studies (Poldrack et al. 2017; Marek et al. 2022; Spisak et al. 2023). Future research should build upon these findings by refining QC measures and incorporating more sophisticated longitudinal models to unravel the temporal dynamics of brain–behavior associations. Expanding this approach to other behavioral domains could further elucidate the interplay between neural development and decision-making processes. In sum, this study underscores the critical role of behavioral data quality in neuroimaging research. Incorporating rigorous QC pipelines, even at the cost of reduced sample size, substantially enhances the magnitude of brain–behavior associations, both in effect sizes and number of statistically significant findings. As large-scale datasets become more common, establishing standardized behavioral QC protocols should be a central priority to ensure the reproducibility and interpretability of findings in developmental cognitive neuroscience.

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Most ABCD research sites rely on a central Institutional Review Board (IRB) at the University of California, San Diego, for the ethical review and approval of the research protocol, with a few sites obtaining local IRB approval. The views expressed in this manuscript are those of the authors and do not necessarily reflect the official views of the National Institutes of Health, the Department of Health and Human Services, the U.S. federal government or ABCD consortium investigators. The views and opinions expressed in this manuscript are those of the authors only and do not necessarily represent the views, official policy or position of the U.S. Department of Health and Human Services or any of its affiliated institutions or agencies.

## Author contributions

Mahmoud Elsayed contributed to conceptualization, data curation, formal analysis, investigation, methodology, software, visualization, writing (original draft), writing (review and editing), project administration, and validation. Wei Q Deng contributed to conceptualization, data curation, formal analysis, investigation, methodology, software, supervision, writing (review and editing), and validation. Kyla L Belisario contributed to data curation, investigation, methodology, resources, validation, and writing (review and editing). James MacKillop contributed to conceptualization, funding acquisition, investigation, methodology, resources, supervision, validation, visualization, writing (original draft), and writing (review and editing).

## Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

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## Conflicts of interest

J.M. is a Principal and Senior Scientist at BEAM Diagnostics, Inc. No other authors have declared any competing interests.

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